Boolean net

Synthesis starts from a technology-independent subject graph

The idea

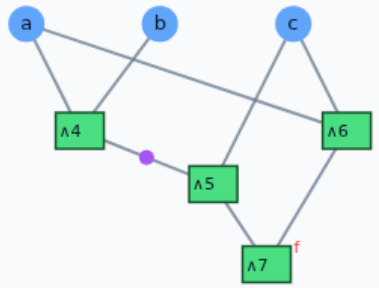
- Our subject graph is an And-Inverter Graph: AND nodes with optional complemented edges

Graph

BOOLEAN NETWORK

Graph

Step 1 / 5 · graph · module01-01-boolean-network



Explanation

Seven nodes: three PIs, four ANDs, one PO f.

- Subject graph
- AIG

```
AND=4
```

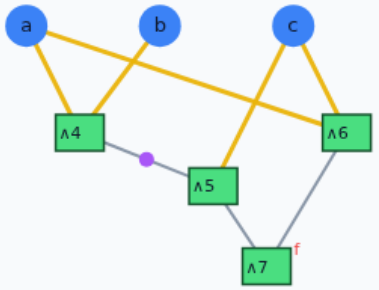
Graph

PIs

BOOLEAN NETWORK

PIs

Step 2 / 5 · pis · module01-01-boolean-network



Explanation

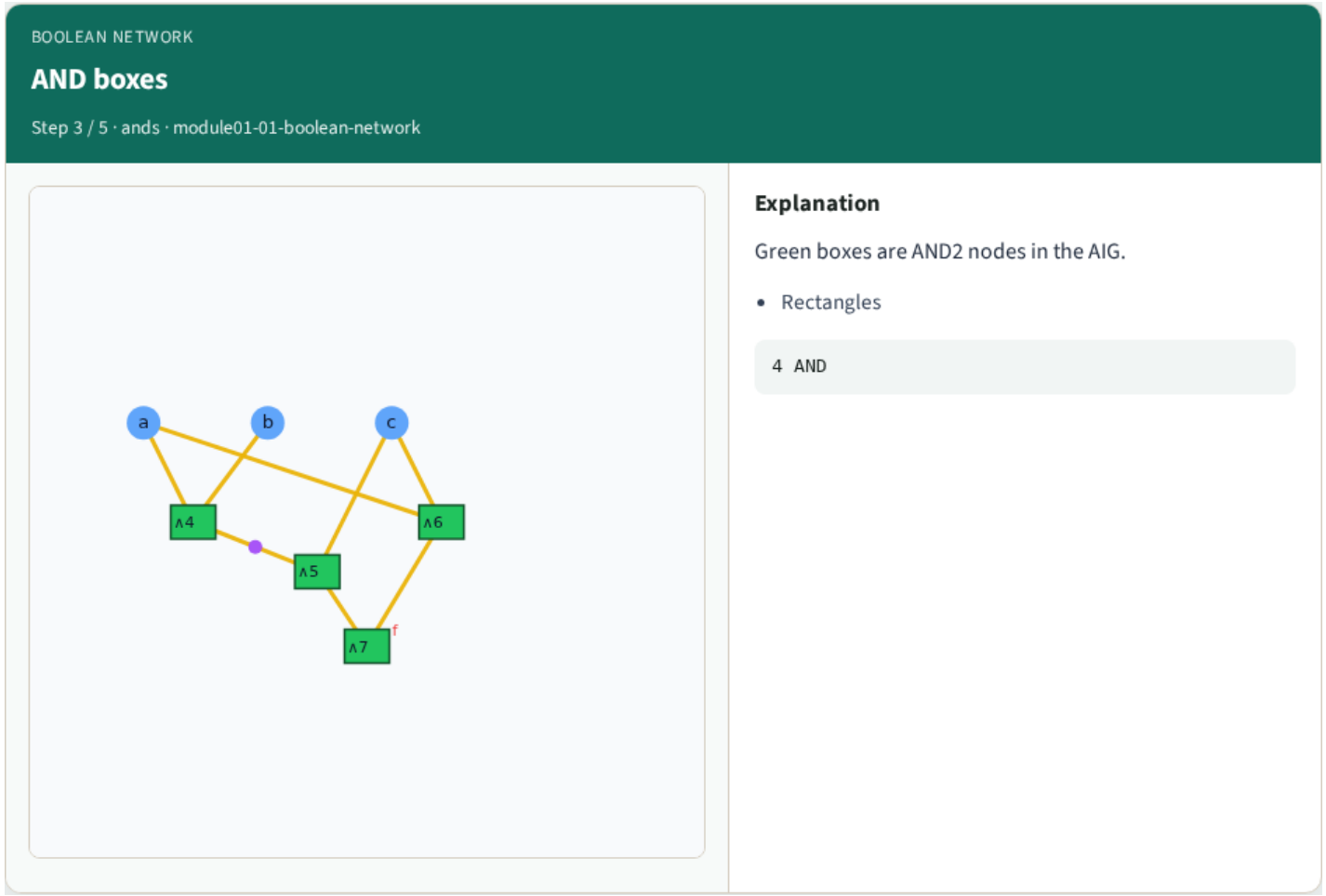
Primary inputs a, b, c sit at the top.

- Circles

```
a b c
```

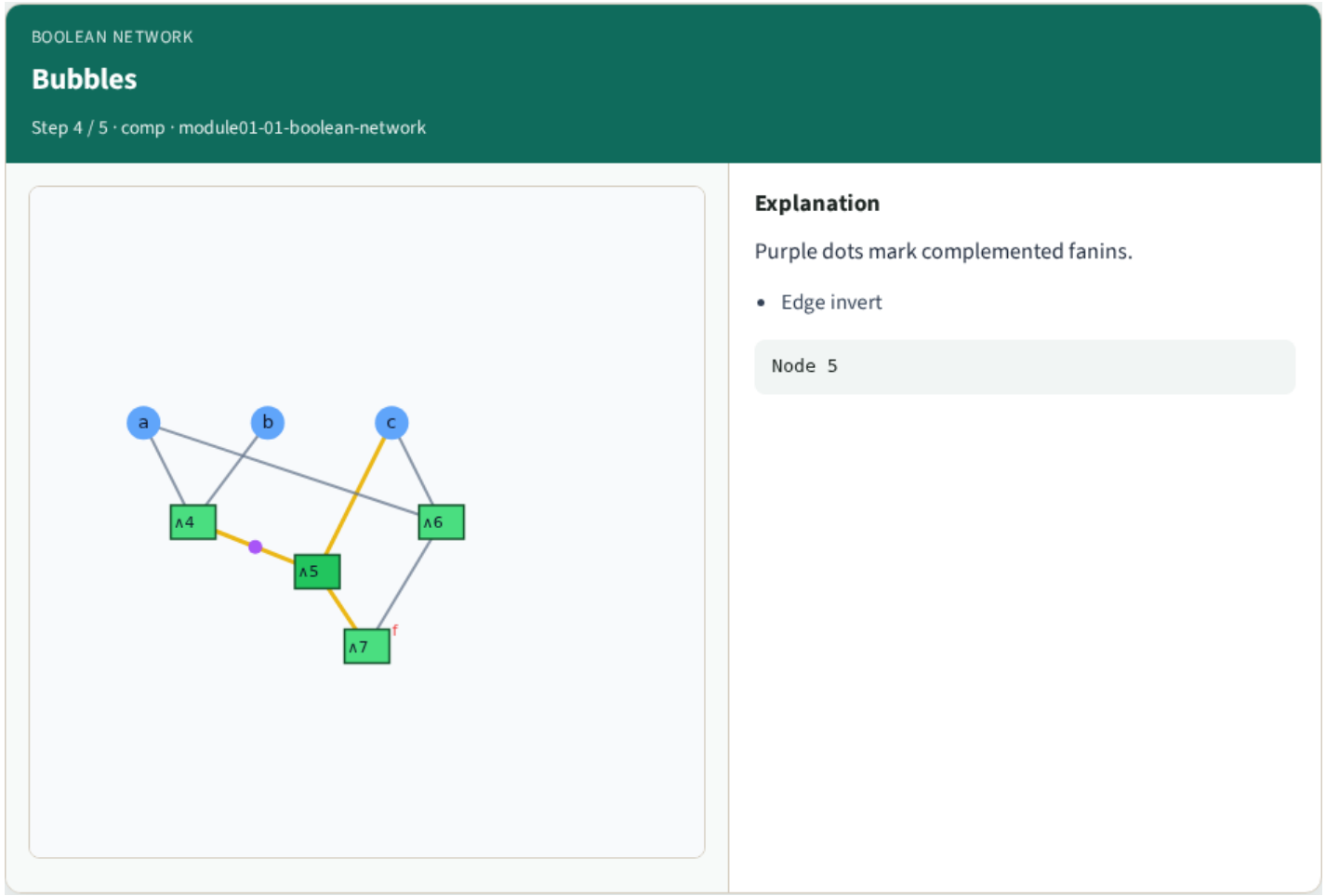
PIs

AND boxes



AND boxes

Bubbles



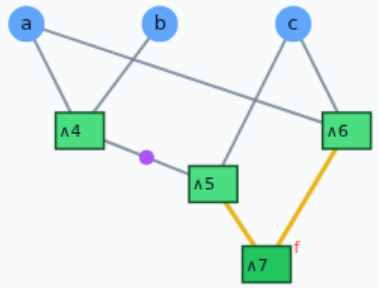
Bubbles

PO

BOOLEAN NETWORK

PO

Step 5 / 5 · po · module01-01-boolean-network



Explanation

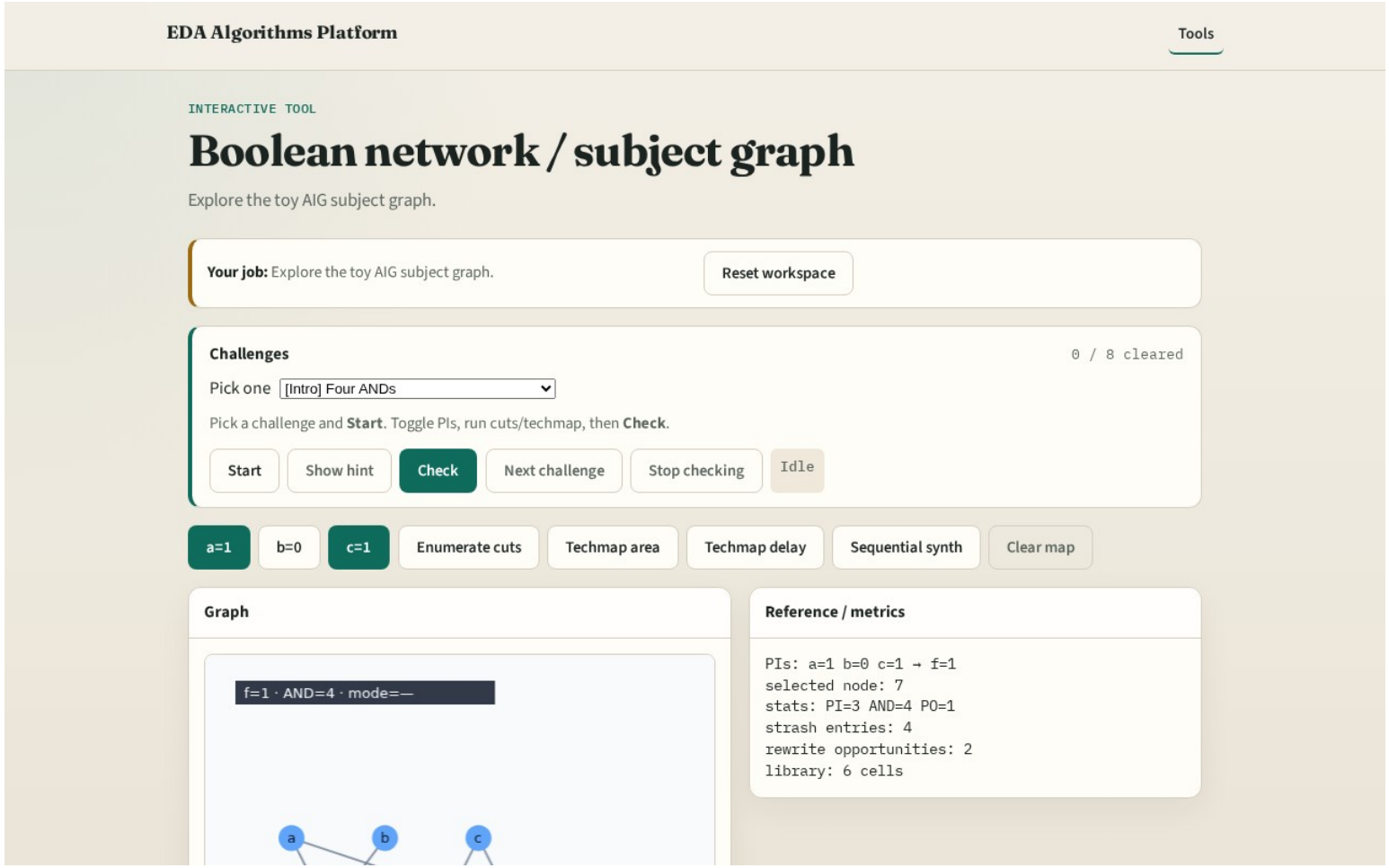
Node 7 drives output f.

- Primary out

f

PO

Browser lab track



Browser lab starter

Implement track

- Load tiny_synth.json; print aig_stats

Pitfalls

- Treating INV as a separate AIG node type
- Ignoring complement bubbles

Your turn

- Next: AIG node types in detail